

Facebook Ads Campaign Analysis & Visualization

Project Overview

This project performs comprehensive analysis and visualization of Facebook advertising campaign data to evaluate marketing effectiveness, spending patterns, and return on marketing investment (ROMI). The analysis combines time-series analysis, campaign performance comparison, distribution analysis, and correlation studies to provide actionable insights into advertising ROI.

Objectives

- **Analyze daily advertising spend and ROMI trends over time**
- **Compare campaign performance across different marketing initiatives**
- **Identify spending patterns and ROI variability**
- **Examine correlations between advertising metrics**
- **Visualize relationships between ad spend and generated value**
- **Calculate and track Return on Marketing Investment (ROMI)**

Tech Stack

- **Programming Language: Python 3.x**
- **Libraries:**
 - **Pandas (data manipulation)**
 - **Matplotlib (visualization)**
 - **Seaborn (statistical visualization)**
 - **NumPy (numerical operations)**
- **Environment: Jupyter Notebook / Google Colab**

- **Data Source:** `facebook_ads_data.csv`

Dataset Structure

The dataset contains Facebook advertising campaign metrics:

Key Fields:

- **ad_date:** Date of advertising activity
- **campaign_name:** Name/identifier of the advertising campaign
- **total_spend:** Total advertising expenditure
- **total_value:** Total value/revenue generated
- **Additional numeric metrics for correlation analysis**

Calculated Metric:

- **ROMI (Return on Marketing Investment):**
$$\frac{(\text{total_value} - \text{total_spend})}{\text{total_spend}} * 100$$

Project Tasks

Task 1: Daily Time-Series Analysis (2021)

Goal: Analyze daily advertising performance throughout 2021

Steps:

1. **Filter data for year 2021 (can use string comparison:**
`ad_date >= '2021-01-01')`
2. **Group data by date**
3. **Calculate daily aggregates**

Visualizations:

- **Chart 1: Daily total advertising spend in 2021**
- **Chart 2: Daily ROMI in 2021**

Bonus Mini-Task:

- **Apply `rolling()` method to display moving averages for both spend and ROMI**

Skills Applied:

- **Date filtering (without explicit date conversion)**
 - **GroupBy operations**
 - **Time-series aggregation**
 - **Line plot creation**
 - **Rolling window calculations**
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Task 2: Campaign-Level Performance Analysis

Goal: Compare performance across different advertising campaigns

Steps:

- 1. Group data by `campaign_name`**
- 2. Calculate aggregate metrics per campaign**

Visualizations:

- **Chart 1: Total advertising spend per campaign**
- **Chart 2: Total ROMI per campaign**

Skills Applied:

- **Categorical grouping**
- **Bar chart creation**
- **Campaign comparison analysis**

Task 3: ROMI Variability Analysis

Goal: Examine the distribution and variability of daily ROMI across campaigns

Visualization:

- **Box Plot:** Daily ROMI distribution by campaign name

Insights Provided:

- Median ROMI per campaign
- Quartile ranges (Q1, Q3)
- Outlier identification
- Campaign consistency/volatility

Skills Applied:

- Box plot creation
- Distribution analysis
- Outlier detection
- Statistical visualization

Task 4: ROMI Distribution Analysis

Goal: Understand the overall distribution of ROMI values

Visualization:

- **Histogram:** Distribution of ROMI values from `facebook_ads_data.csv`

Insights Provided:

- ROMI frequency distribution
- Central tendency visualization

- Spread and skewness identification

Skills Applied:

- Histogram creation
 - Distribution shape analysis
 - Bin optimization
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Task 5: Correlation Heatmap

Goal: Identify relationships between all numeric metrics

Visualization:

- Heatmap: Correlation matrix of all numeric variables

Analysis Questions:

- Which metrics have the highest correlation?
- Which metrics have the lowest correlation?
- What correlates with `total_value`?

Skills Applied:

- Correlation matrix calculation
 - Heatmap visualization with Seaborn
 - Feature relationship analysis
 - Color mapping for correlation strength
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Task 6: Spend vs. Value Relationship

Goal: Visualize and model the relationship between advertising spend and generated value

Visualization:

- **Scatter Plot with Linear Regression: Using `lmplot()` function**
 - **X-axis: `total_spend`**
 - **Y-axis: `total_value`**

Insights Provided:

- **Linear relationship strength**
- **Regression line (trend)**
- **Confidence intervals**
- **ROI efficiency visualization**

Skills Applied:

- **Scatter plot creation**
- **Linear regression modeling**
- **Seaborn `lmplot` usage**
- **Relationship visualization**

Key Deliverables

A Jupyter Notebook (.ipynb) or Google Colab link containing:

Required Components:

1. Data Loading

- **Successfully loaded `facebook_ads_data.csv` into Pandas DataFrame**
- **Data validation and initial exploration**

2. Task 1 Outputs

- **2021 data grouping by date**
- **Daily ad spend line chart (2021)**
- **Daily ROMI line chart (2021)**

- **(Bonus) Moving average overlays**
 - 3. Task 2 Outputs**
 - **Campaign-level data grouping**
 - **Total spend by campaign bar chart**
 - **Total ROMI by campaign bar chart**
 - 4. Task 3 Output**
 - **Box plot showing daily ROMI distribution per campaign**
 - 5. Task 4 Output**
 - **ROMI distribution histogram**
 - 6. Task 5 Output**
 - **Correlation heatmap with annotations**
 - **Written analysis of:**
 - **Highest correlation pairs**
 - **Lowest correlation pairs**
 - **total_value correlations**
 - 7. Task 6 Output**
 - **Scatter plot with regression line (total_spend vs total_value)**
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Skills Demonstrated

- **Data Manipulation: Filtering, grouping, aggregation**
- **Time-Series Analysis: Date filtering, rolling windows, trend analysis**
- **Statistical Visualization: Histograms, box plots, correlation matrices**
- **Regression Analysis: Linear modeling, relationship visualization**
- **Matplotlib/Seaborn Proficiency: Multiple chart types, styling, annotations**
- **Business Metrics: ROMI calculation, ROI analysis**

- **Comparative Analysis:** Campaign performance evaluation
 - **Data Storytelling:** Translating visualizations into insights
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Analysis Focus Areas

This project examines:

- **Temporal Patterns:** Daily spending and ROI trends
- **Campaign Effectiveness:** Which campaigns deliver best ROMI
- **Investment Efficiency:** Relationship between spend and returns
- **Performance Consistency:** ROMI variability across campaigns
- **Metric Relationships:** How advertising metrics correlate
- **ROI Optimization:** Identifying high-performing investment levels
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